

Final Paper

A Crowdsourced Accessibility Mapping Platform

Introduction

Accessibility in public spaces remains an ongoing challenge for individuals with disabilities. Despite legislation such as the Americans with Disabilities Act (ADA), many communities still lack accessible infrastructure. There is no easy, accessible record of which public areas pose challenges. My proposed open-source project which I call AccessMap, will aim to address this gap by creating a crowdsourced accessibility mapping platform. The platform will show users ways to report, view, and talk about accessibility barriers in real-time. It will build a continuously updated map for public use. This paper outlines the goals, challenges, and open-source management strategies for developing AccessMap over a projected five year period timeline. With open source and working principles like transparency, community collaboration, inclusivity, and shared governance. The project aims to not just build a tool but to create a community dedicated to accessibility awareness and improvement.

Project Description

AccessMap will function as a mobile and web-based platform where users can submit reports about accessibility barriers, for example broken ramps, blocked sidewalks or non-functioning elevators. Users will be able to view existing reports shown on top of an interactive map using geographic information systems (GIS). There will be a community that users can interact with like by commenting and having an upvote or downvote system to validate the reports. They can even have an ability to discuss potential solutions or updates. The audience attending this includes more than people with disabilities. It includes caregivers, urban planners, local government officials, and the local community in general. The crowdsourcing reports will help empower the overall community to document and advocate for accessible spaces.

The specific details of the project and how it will work for a user is that the user will make reports using descriptions and location of where they are reporting with the added option of uploading photos. There will be moderation tools for reviewing and verifying these reports. With these verified reports it will become interactive on the maps and have filtering options. API access for third-party integration with civic technology apps will be considered as well. Discussion forums for the community will be there for dialogue and recommendations.

Project Management Using Open Source Principles

Managing an open source project requires more than releasing the source code publicly. It requires embedding collaboration, transparency, and community ownership into every process. I will describe how I would apply open source and working open principles to AccessMap. Firstly, transparency and open communication is important. All development will occur in public repositories like GitHub with issue tracking, pull requests, and roadmaps accessible to contributors and users alike. Major decisions, whether it is technical, governance, or design-related. It will be documented in project wikis and discussed in open community meetings. Frequent community meetings and forums will invite feedback, explain current challenges and maintain open channels for reporting issues that goes beyond the software, like organization bottlenecks.

Community collaboration will be a main focus. I would publish clear contributor guidelines that show how to submit bug reports, suggest features or contribute code. There would also be a code of conduct to ensure that the space is respectful and inclusive. A core maintainer team will help manage contributions, mentor newcomers, and make sure that the code is good quality. I want to bring in people who have experience with accessibility, either through their profession or lived experience. That way the platform will reflect the needs of the people using it. We will also use shared governance which means that I won't be the only one making decisions. Core maintainers will handle technical choices, but for big priorities or changes, the community will get to vote. I want to have an advisory council of accessibility experts, advocates, and civic tech people to help guide us and make sure we're staying ethical and fair.

Managing Technical Debt There will be technical debt in this kind of project. With so many features and functions, bugs or slow code can build up. To help with this, I will use automated testing early in development. This will catch problems faster and stop them from growing later. All code will also go through peer review before it is added to the main project. Having structured design documentation will also help new developers understand how the platform works. Every six months we will do a technical debt audit where we go through the code and check what needs to be updated, removed, or fixed. We will keep a public tracker of known technical debt so that contributors can work on it whenever possible.

5 Year plan

Five-Year Development Plan The development of AccessMap will happen over five phases. The first phase is the foundation that will take about 6 months. In this phase, we define the scope and features, set the tech stack, and design the UI and database. Internal testing will happen and we will deploy an MVP for early feedback. The second phase will be beta testing and more refining that will take place during the 6 months after the first completing

the first year. A beta version will be launched in a few areas. We will get user feedback and make improvements. Knowledge-sharing and early technical debt tracking will begin the third which will bring us scaling up the project during the second year. The infrastructure will be improved and we will introduce cloud storage, and focus on performance and security. More testing and training will happen. The fourth phase will be during the third year and will be helping with the stability and optimization of the app. We keep everything up to date, improve security, fine-tune performance for high traffic, and roll out major UX upgrades. The last phase will be during the last 2 years of the five year plan and this will be us futureproofing and increase the growth of the platform. This includes developing new features based on user needs, moving to microservices, finding funding, and planning for bigger upgrades or even a system overhaul.

Automation in Platform Management

Automation and Community Automation can help a lot, especially with large amounts of user input. It can help flag duplicates, approve simple reports, and manage moderation queues. That keeps the platform responsive even when things pile up. It can also track technical debt, alert us when deadlines are near, or point out security risks. But automation has to be used wisely. If replies feel too robotic, users might stop engaging. Also, automation can't catch everything. Human moderators will still be needed for complex issues and mentoring new users. Especially early on, too much automation might slow us down. It should be added gradually and used where it makes the most difference.

Anticipated Challenges

Anticipated Challenges As the project grows, there will be more things to manage. One of them is technical debt keeping our software, tools, and dependencies up to date. As more users join and the database grows, it will become harder to maintain performance. Cloud services and extra computing power will be necessary. The team will also get bigger and new people might not be familiar with the older parts of the system. This can lead to mistakes or inconsistent code. To reduce these issues, we need to have regular documentation updates and knowledge-sharing sessions.

Another challenge is contributor engagement. Open-source projects sometimes struggle with burnout or people leaving. We will focus on mentorship, shared ownership, and good communication. By involving the community and spreading responsibility, we make the project more sustainable.

Conclusion

AccessMap is not just a platform, it's a community driven effort to make the world more accessible. By applying open source and working open principles, we can build a project

that empowers people, stays sustainable, and evolves. With the right planning, collaboration, and commitment to accessibility, AccessMap can become a powerful tool for public advocacy and inclusive design.

Sources:

Lecture Slides

<https://opensource.org/osd>